

From EHR to evidence

A visual guide to the MIMIC-IV-ED dataset exploration notebook

What students will learn

- How an emergency-department stay becomes one analysis row
- How to read variables, distributions, frequencies, and missingness
- How imputation and correlation support, but do not replace, judgment

A notebook is a chain of questions

01

Context

What is an EHR, and why does it matter?

02

Construction

How did MIMIC-IV-ED become this table?

03

Inspection

What do the fields, distributions, and gaps say?

04

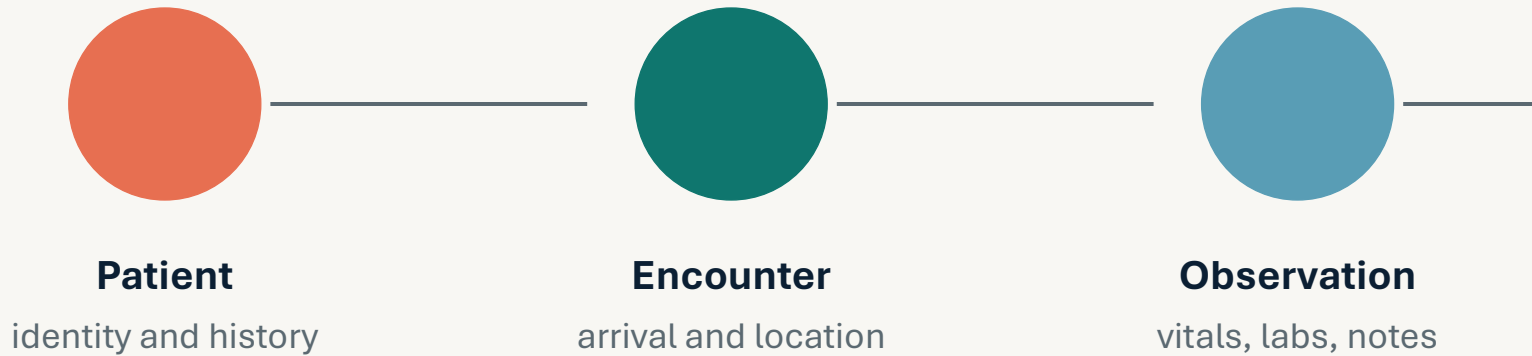
Preparation

What is cleaned, imputed, audited, and exported?

The habit: every number should have a definition, a time window, and a provenance.

What is an electronic health record?

An EHR is a longitudinal digital record of care: observations, actions, decisions, and outcomes captured across clinical workflows.



Why the structure matters

- The same person can have many encounters. A value has meaning only with time and units.
- Documentation reflects workflow, not just physiology.
- Labels can be defined after the encounter ends.

Why EHR data is powerful - and difficult

Scale

Many encounters and measurements make patterns visible.

Realism

Data reflects actual workflows, missingness, and operational choices.

Timing

The same value can be valid for description but invalid for triage prediction.

Responsibility

Privacy, bias, leakage, and clinical interpretation stay in scope.

The data is valuable because it is close to care. That is also why definitions matter.

MIMIC-IV-ED: a clinical data resource

MIMIC is a large, de-identified database designed for reproducible clinical research and education.

ED

emergency-department
encounters

IV

major MIMIC data release

ED stay

the row grain used in M1

MIMIC context is important, but the notebook's immediate object is the locally extracted master table.

M1 notebook

- Uses the extracted teaching table.
- Recomputes claims from the local generated CSV.
- Treats the table as a research artifact, not a clinical tool.
- Keeps provenance and limitations visible.

How the dataset becomes one analysis table

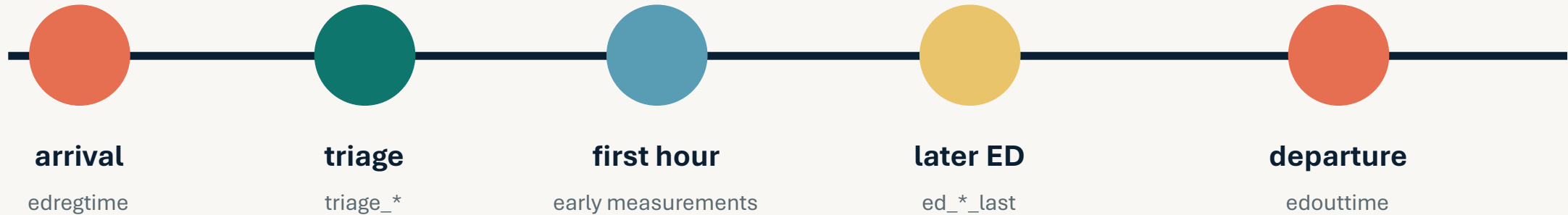


Join logic is part of the data. A row is a constructed claim about an encounter.

The notebook verifies row counts, unique stay_id values, repeat subjects, column families, and timing boundaries before analysis.

Read every variable against the encounter timeline

One row = one ED stay



Availability is a modeling decision: a later ED measurement may be descriptive but outside a triage-time prediction boundary.

One subject can have multiple stays

The variable dictionary is a map of meaning

Identifiers & timing

stay_id, subject_id, hadm_id, intime, outtime

Demographics & arrival

age, gender, race, insurance, transport

Triage measurements

temperature, heart rate, oxygen saturation, acuity

Prior utilization

ED, hospital, and ICU counts over 30/90/365 days

Later ED measurements

ed_*_last and medication counts

Outcomes

LOS, hospitalization, critical outcome, revisit

A plausible name is not a definition. Read the generated dictionary and the extraction contract.

How to read the symbols and suffixes

NaN

missing value; not
automatically zero or normal

***_last**

last recorded ED measurement
in the defined window

**n*_30d / 90d /
365d**

count over a prior utilization
window

outcome_*

label or downstream event,
not a triage observation

_iterative

a new imputed analysis
column; raw field is preserved

%

a proportion expressed per
100 rows

The same column name carries both a data type and a time meaning.

Summary statistics are compact descriptions, not verdicts

Mean

arithmetic average; sensitive to extreme values

$\text{sum}(x) / n$

Standard deviation

typical spread around the mean

$\text{sqrt}(\text{variance})$

Median

middle ordered value; robust to skew

50th percentile

IQR

middle 50% of the distribution

75th - 25th percentile

Percentiles

locate values in the ordered distribution

p01, p25, p50, p75, p99

Missing %

how much of a column is absent

$\text{missing} / \text{total} * 100$

For skewed clinical variables, show the median and IQR alongside the mean and standard deviation.

What the current M1 table says at a glance

425,087

ED-stay rows

120

variables in the CSV

205,504

unique subjects

69,952

subjects with multiple stays

Variable	Median	IQR	99th pct	Missing %
age	53.00	34.00	93.00	0.02%
triage_acuity	3.00	1.00	4.00	1.64%
triage_temperature	98.00	1.10	101.30	5.51%
triage_hearttrate	84.00	24.00	134.00	4.02%
triage_o2sat	99.00	3.00	100.00	4.85%
triage_sbp	133.00	28.00	198.00	4.30%
ed_los_hours	5.47	4.79	30.48	0.00%

Interpretation requires units: temperature is stored in Fahrenheit-like values when detected; ED LOS is in hours; acuity is ordinal 1-5.

Categorical frequencies describe representation

A frequency table answers: which labels appear, how often, and how much of the table do they cover?

triage_acuity

ordinal

3.0: 225,066 | 2.0: 139,411 | 4.0: 28,504

gender

nominal

F: 229,898 | M: 195,189

race

nominal

WHITE: 228,123 | BLACK/AFRICAN AMERICAN: 76,798
OTHER: 20,752

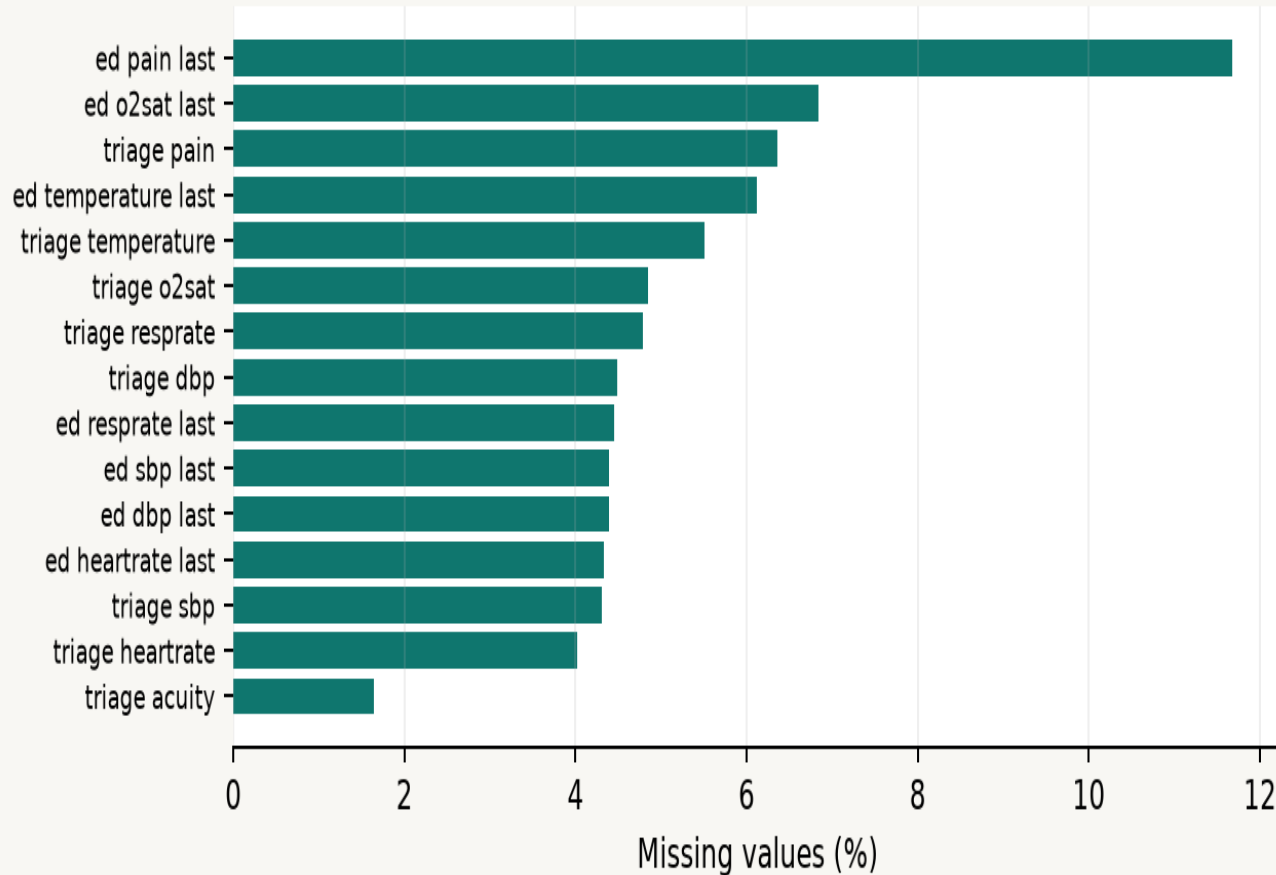
arrival_transport

nominal

WALK IN: 251,849 | AMBULANCE: 155,752 |
UNKNOWN: 15,352

Frequency is not causality, quality, fairness, or clinical meaning. Rare and missing categories need review before encoding.

Missingness is part of the data-generating process



A blank can mean

- the test was not ordered
- the value was not documented
- the event fell outside the time window
- the extraction did not find a match
- the measurement was removed as implausible

M1 inspects missingness before choosing an imputation strategy.

Imputation: estimate carefully, validate honestly

The notebook creates a known test by hiding 20% of cells in rows that were originally complete.



Compared methods

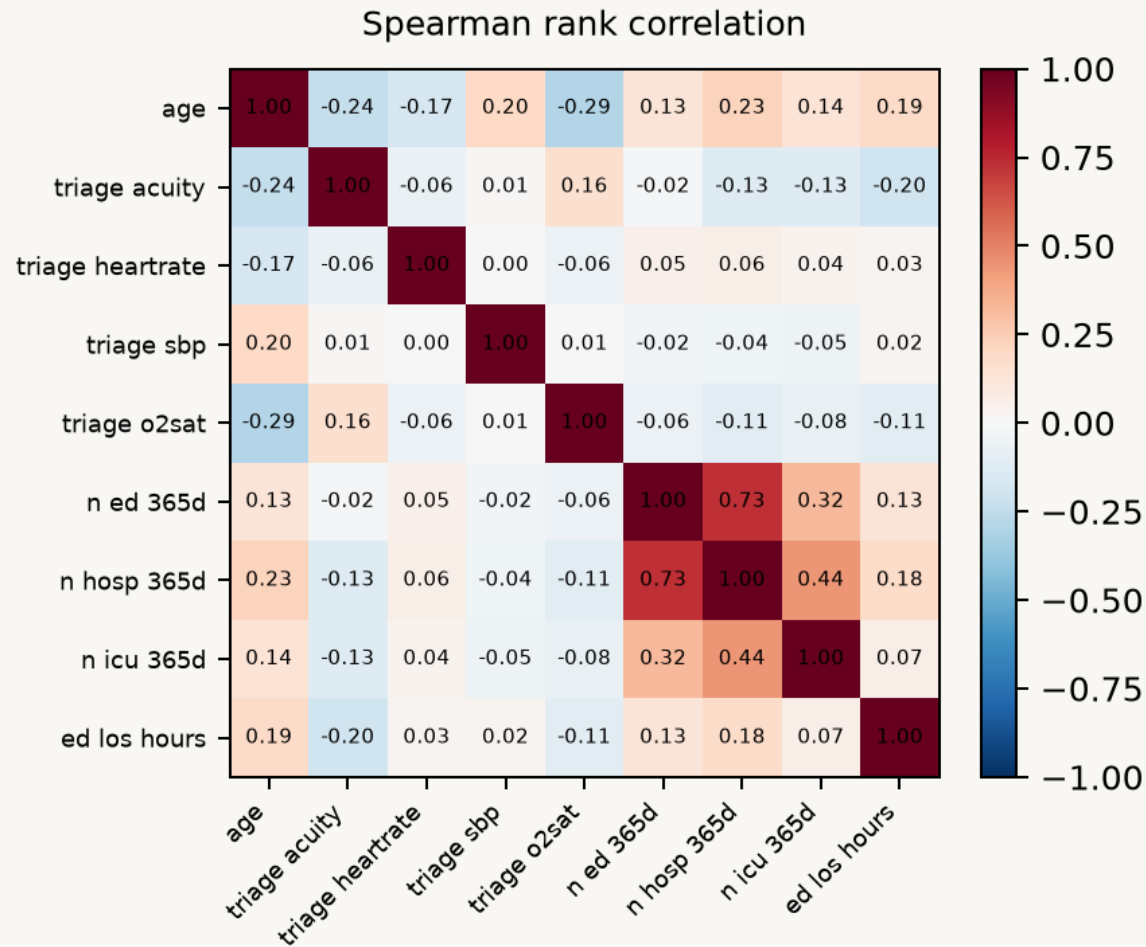
Median | KNN | Iterative | Autoencoder-style MLP

Important guardrails

Range-clean first | constrain outputs | score masked cells only | preserve raw data

Local result: iterative selected | validation rows: 10,000 | masked cells: 29,943

Correlation asks whether variables move together



How to read it

- Spearman correlation uses ranks, so it captures monotonic association.
- +1 means variables rise together; -1 means one rises as the other falls.
- 0 means little monotonic association in this table.
- Correlation is not causation and does not prove predictive usefulness.
- Different time windows can create leakage concerns even when correlation is strong.

What M1 hands forward

M1_variable_dictionary.csv

families, dtypes, non-missing counts, missing percentage

M1_missingness.csv

column-level missingness for audit and discussion

M1_imputation_comparison.csv

MSE, MAE, and masked-cell counts by method

M1_vital_imputation_audit.csv

raw vs imputed names, ranges, and missing-after checks

M1_quality_checks.csv

negative LOS, zero LOS, IDs, targets, and range flags

M1_target_summary.csv / M1_acuity_summary.csv

outcome prevalence and descriptive acuity comparisons

Raw vital columns stay available; selected method-suffixed columns provide complete modeling estimates.

The student workflow inside M1

- 1 Run from the workspace root or confirm the data path.
- 2 Verify rows, columns, unique stays, and repeated subjects.
- 3 Read the generated variable dictionary before plotting.
- 4 Compare numerical summaries with units and missingness.
- 5 Inspect categorical frequencies without hiding missing labels.
- 6 Compare imputation methods on artificially masked cells.
- 7 Review leakage boundaries before using a field as a predictor.
- 8 Inspect the exported handoff tables and document limitations.

The deliverable is not just a cleaned table. It is a defensible explanation of what the table means.

Before you model, be able to say what the row means

A strong analysis begins with four answers:



Which people and encounters are represented?



What does one row and each variable represent?



When could each value have been known?



What is observed, missing, cleaned, imputed, or downstream?

M1 turns a clinical data extract into a transparent analytical starting point.